Journal of Sustainability for Energy
https://www.acadlore.com/journals/JSE

Energy Renovation of a Hospital Support Building for Sustainable Energy Use: Scenario Evaluation and Investment Trade-Offs

Tatjana Ružičić[1], Srđan Vasković[2*], Milovan Kotur[3], Jovan Škundrić[3], Djordje Vojinovic[4], Gojko Krunić[5]

[1] University Clinical Center of the Republic of Srpska, Technical-Economic Block (TEB), 78000 Banja Luka, Bosnia and Herzegovina
[2] Faculty of Mechanical Engineering, University of East Sarajevo, 71123 East Sarajevo, Bosnia and Herzegovina
[3] Faculty of Mechanical Engineering, University of Banja Luka, 78000 Banja Luka, Bosnia and Herzegovina
[4] Faculty of Technology, University of Banja Luka, 78000 Banja Luka, Bosnia and Herzegovina
[5] Faculty of Production and Management, University of East Sarajevo, 89101 Trebinje, Bosnia and Herzegovina

[*] Correspondence: Srđan Vasković (srdjan.vaskovic@ues.rs.ba)

Received: 06-12-2026 **Revised:** 07-23-2026 **Accepted:** 07-30-2026

Citation: T. Ružičić, S. Vasković, M. Kotur, J. Škundrić, D. Vojinovic, and G. Krunić, "Energy renovation of a hospital support building for sustainable energy use: Scenario evaluation and investment trade-offs," *J. Sustain. Energy*, vol. 5, no. 3, pp. 195–212, 2026. https://doi.org/10.56578/jse050302.

© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Renovating hospital support buildings requires balancing reductions in fossil-fuel use against capital expenditure and operational constraints. This study examines the energy and investment trade-offs among renovation scenarios for an existing hospital support building. An annual energy model was reconstructed from available engineering estimates for the Technical-Economic Block (TEB) of the University Clinical Centre of the Republic of Srpska in Banja Luka. Envelope renovation, boiler replacement and solar thermal installation were combined into seven scenarios. Primary-energy savings, operational CO_2 reductions, investment and fuel-cost savings were compared. Preference Selection Index (PSI) rankings were calculated using four criteria and a reduced two-criterion formulation, followed by an analysis of alternative investment weights. The three-measure package achieved the largest estimated primary-energy savings, at 1.813 GWh/year, and ranked first under both PSI formulations. However, boiler replacement alone ranked first when energy savings and investment received equal weights. Envelope renovation combined with boiler replacement retained 98.1% of the three-measure package's energy savings at 12.9% lower investment. Adding solar thermal required a further 322,337 Bosnia and Herzegovina convertible marks (BAMs) and produced estimated annual fuel-cost savings of 4,409 BAM, giving an incremental simple payback of 73.1 years. Estimated operational CO_2 reductions reached 454.5 tonnes/year using a conversion factor inferred from the source scenario data. These findings distinguish maximum energy savings from investment preference and marginal financial returns. The assessment supports preliminary renovation comparisons, while its absolute estimates remain conditional on verification of the baseline and engineering inputs.

Keywords: Sustainable energy use; Hospital energy renovation; Thermal-envelope retrofit; Boiler replacement; Solar thermal energy; Multi-criteria decision making; Incremental economic analysis

1 Introduction

Renovating existing buildings can reduce operational energy demand while extending the use of buildings that remain functionally adequate. The challenge is to identify interventions that deliver worthwhile reductions in energy consumption and emissions within the financial and technical constraints of each project. A review of energy retrofitting at the multi-building scale highlights the need to give greater attention to non-residential buildings and to the conditions under which individual retrofit solutions can be transferred to larger building portfolios [1]. Research on multi-objective retrofit optimisation has also moved towards combining energy simulation, economic assessment and environmental indicators, although the selection of an implementable solution remains sensitive to model assumptions and project priorities [2]. Climate-adaptation research further shows that a renovation strategy should be judged against the conditions in which the building will operate, rather than solely against its present energy demand [3]. These considerations make the comparison of renovation packages a necessary part of sustainable energy planning.

Hospitals present particular difficulties because reductions in energy use must remain compatible with service continuity, indoor environmental requirements and reliable heat supply. Their energy demand arises from a combination of clinical activities, ventilation, domestic hot water and support services, with substantial differences between buildings and functional areas. Psillaki et al. [4] identify technological choices, resource management and organizational practices as interrelated influences on hospital energy performance. A review of hospital retrofitting for indoor air quality and energy efficiency likewise shows why energy reduction cannot be considered independently of the conditions required by building users [5]. At the institutional level, an assessment of the net-zero strategies of 214 National Health Service trusts identifies shortcomings in monitoring and performance measurement, despite widespread commitments to decarbonization [6]. Together, these studies indicate that credible hospital renovation decisions require a clearly defined assessment boundary and measurable outcomes, as well as an explicit account of the constraints under which measures can be implemented.

Within this broader hospital context, renovation studies increasingly examine several objectives simultaneously. Shi et al. [7] developed a decision-making approach for hospital wards that considered energy performance, economic viability and thermal comfort, making investor preferences part of the selection process. Subsequent work combined automated machine learning with multi-objective optimization to investigate hospital retrofitting under changing climatic conditions [8]. These approaches demonstrate the value of exploring alternatives rather than selecting measures on energy savings alone. Their detailed treatment of hospital performance also underlines the importance of matching the assessment to the function of the building. A support building containing kitchens, laundries and technical services has a different demand structure from a patient ward; consequently, neither its energy balance nor its renovation priorities should be inferred directly from ward-based assessments.

The relationship between building demand and heat supply is especially important when envelope renovation and equipment replacement are considered together. A review of heating, ventilation, air-conditioning and domestic hot-water retrofits identifies heat generation and system control as recurring areas of intervention [9]. However, a supply-system assessment must reflect the demand that remains after the proposed building works. Envelope improvements alter both annual heat requirements and the load conditions under which replacement equipment operates. An evaluation of heating alternatives for industrial buildings similarly shows that system selection depends on economic, spatial and logistical considerations in addition to energy performance [10]. Domestic hot-water demand requires separate attention because its temperature and hygiene requirements do not necessarily change with space-heating demand. Comparative simulations of hot-water systems in well-insulated residential buildings illustrate the interaction between efficiency and hygiene constraints [11]. Although residential results cannot be transferred directly to hospitals, the distinction between space-heating and hot-water requirements remains relevant to defining the energy balance of a hospital support building.

Solar thermal integration introduces a further interaction between renewable heat production and the conventional system it displaces. An energy audit of a hospital hot-water installation evaluated changes to an existing solar-and-boiler system with the aim of increasing the solar contribution [12]. Matera et al. [13] investigated a different configuration, in which a compound parabolic concentrator supplied two hot-water uses at different temperatures through separate storage arrangements. Research on robust hospital solar-system sizing also shows that predicted performance depends on uncertainty in climatic conditions, demand and system operation [14]. These studies establish the relevance of solar heat to hospital energy supply, but they do not imply that a given collector installation is economically attractive in every renovation package. For a fixed useful solar output, the amount of fuel displaced depends on the efficiency of the auxiliary heat source. A solar installation evaluated alongside an old boiler may therefore have a different financial return when it is added after boiler replacement.

This dependence makes the economic comparison of renovation packages more demanding than a comparison of their total savings. A cost analysis of residential retrofits in Bosnia and Herzegovina found substantial differences between building categories, illustrating the importance of retaining the characteristics of the assessed stock when interpreting investment requirements [15]. A recent review of financial evaluation in building retrofits identifies life-cycle costing and cost–benefit analysis as important assessment approaches, while also documenting inconsistent treatment of economic assumptions [16]. Simple payback remains useful as an initial indicator, but it does not account for financing, equipment replacement, residual value or the timing of expenditure and savings. It also answers a different question from incremental analysis. The payback of an entire renovation package describes its aggregate investment and savings; it does not establish whether an additional measure provides an attractive return relative to a less expensive package.

The environmental boundary requires comparable care. Recent renovation decision frameworks incorporate environmental, economic and social life-cycle metrics to evaluate consequences beyond operational energy use [17]. Research combining life-cycle assessment with multi-criteria decision-making (MCDM) further distinguishes energy and emissions from resource use and circularity when selecting building components [18]. Reviews of dynamic life-cycle assessment and uncertainty treatment show that environmental results depend on temporal assumptions, data quality and the stages included in the assessment [19, 20]. These findings are relevant to the interpretation

of operational retrofit studies. Reductions in fuel use and associated CO_2 emissions provide evidence of improved operational performance, but they do not establish the life-cycle sustainability of an intervention. Where embodied impacts and replacement cycles are not quantified, the conclusions should remain within the operational energy and investment boundary actually assessed.

MCDM provides a way to compare these different consequences without reducing the decision to a single physical or financial indicator. A review of building assessment and retrofitting applications documents the use of MCDM to combine environmental, economic, social and safety considerations [21]. In residential renovation, pairwise comparison and the Technique for Order Preference by Similarity to Ideal Solution have been used to connect stakeholder objectives with the selection of alternatives [22]. Interactive building-performance assessment has also combined sensitivity analysis with the Analytic Hierarchy Process to make the implications of design preferences more accessible [23]. In school retrofit planning, a framework incorporating energy performance, health, educational outcomes and costs found that the preferred intervention depended on both stakeholder preferences and the alternatives considered [24]. The common lesson is that a ranking is conditional on the decision model: the choice of criteria and alternatives helps determine what the resulting preference score represents.

Accordingly, the reliability of a renovation recommendation requires attention to how the ranking was obtained. A robust framework for thermal-insulation selection explicitly examines conflicting stakeholder interests [25], while a comparison of MCDM methods for passive-design selection investigates changes in rankings under varying criterion weights [26]. Retrofit optimisation research has also incorporated different decision-making priorities when interpreting Pareto solutions under future climate conditions [27]. At the modelling level, studies of construction and occupant-behaviour uncertainty show how uncertain inputs can affect energy and comfort outcomes [28], and a recent review identifies climate uncertainty as a continuing concern in retrofit analysis [29]. These are distinct sources of uncertainty. Weight sensitivity examines the consequences of changing decision priorities, whereas uncertainty in heat demand, system efficiency or solar yield affects the performance estimates being ranked. Agreement within one ranking exercise should not be interpreted as validation of the underlying engineering model.

The Preference Selection Index (PSI) method offers a comparatively compact procedure for normalising criteria, deriving weights from the decision matrix and aggregating alternative performance. It was introduced for material selection by Maniya and Bhatt [30] and subsequently applied to decisions arising during production-system design [31]. Its use in hospital renovation is already established: Kadric et al. [32] evaluated renovation measures for hospital buildings in Bosnia and Herzegovina using energy, environmental and financial criteria. Building on this application, the present study examines how interactions between renovation measures, dependence among evaluation criteria and incremental investment returns affect the interpretation of scenario rankings. PSI provides the reproducible reference ranking against which these engineering and investment trade-offs are examined.

An additional consideration arises when several criteria are calculated from the same underlying energy saving. Under a fixed fuel type, emission factor and fuel price, primary-energy savings, avoided CO_2 emissions and fuel-cost savings can be proportional, or nearly proportional, across the alternatives. Although these quantities are useful for communicating different consequences of renovation, including all three in an additive preference model may repeatedly represent the same physical change. Data-derived weights do not remove this dependence or establish that the resulting allocation reflects institutional priorities. For the present application, the decision problem is therefore not resolved by identifying the highest PSI score. It also requires examining whether the selected criteria provide independent information and whether the additional savings of a more comprehensive package justify its additional investment.

This study evaluates seven renovation scenarios for the Technical-Economic Block (TEB) of the University Clinical Centre of the Republic of Srpska in Banja Luka. The scenarios comprise individual and combined applications of thermal-envelope renovation, boiler replacement and solar thermal installation for domestic hot water. The assessment addresses the space-heating and hot-water energy uses defined by the available engineering records, rather than the complete energy demand of the hospital complex. Annual primary-energy savings, operational CO_2 reductions, investment costs and fuel-cost savings are evaluated with explicit consideration of the interaction between reduced heat demand and heating-system efficiency. PSI is used to establish a reference ranking, and the interpretation of that ranking is examined alongside criterion dependence, alternative weighting assumptions and the incremental economics of adding solar thermal to the envelope-and-boiler package.

The study addresses two related questions: which package provides the largest estimated operational energy reduction within the assessed options, and how much additional investment is required to obtain the savings beyond those delivered by a less expensive package? Its contribution is an interaction-aware comparison of renovation choices for a hospital support building, with particular attention to the distinction between aggregate performance and marginal investment returns. By making the engineering assumptions and decision-model dependencies explicit, the assessment supports the interpretation of complete and phased renovation options under limited capital availability. The resulting recommendations remain specific to the analysed building, measures and price assumptions; the transferable element is the evaluation procedure and the evidence required to justify a choice.

2 Case Study and Baseline Energy Assessment

2.1 Building Characteristics and Assessment Boundary

The case study concerns the TEB of the University Clinical Centre of the Republic of Srpska in Banja Luka, Bosnia and Herzegovina. Located in the northern part of the hospital complex, the building accommodates the central kitchen, laundry, maintenance facilities, storage areas and administrative functions. It also houses infrastructure serving other hospital buildings, including heat, steam, water, cooling, compressed-air and medical-gas services. The building and energy-system characteristics used in this assessment were taken from the project information from the original feasibility study.

Constructed in 1976 on sloping terrain, the TEB comprises two basement levels and a ground floor, with maximum plan dimensions of approximately 168 m × 36 m. Its reported usable floor area is 12,947.28 m^2, the heated volume is 62,434.72 m^3, and the external-envelope area is 17,113.65 m^2. Figure 1 shows the building. For the annual energy assessment, the conditioned spaces were represented as a single thermal zone with an indoor design temperature of 22 ℃ and a continuous operating schedule. These assumptions provide a common basis for comparing the renovation scenarios; they do not represent the temperature requirements or operating profiles of every room.



Figure 1. TEB of the University Clinical Centre of the Republic of Srpska in Banja Luka

Note: TEB = Technical-Economic Block.

The assessment covers the space-heating and domestic hot-water demands assigned to the TEB in the engineering calculations. It does not represent the complete energy balance of the hospital complex. Electricity consumption, cooling, medical equipment, steam production and other process loads were not added to the assessed heat demand. Consequently, the calculated savings describe changes within the defined heating and hot-water boundary, rather than reductions in total hospital energy consumption.

2.2 Existing Heat Supply and Operating Assumptions

The existing heat supply is associated with a central boiler plant. The original assessment identifies a steam boilers factory (tvornica parnih kotlova, TPK) Zagreb BKG 60a fuel-oil hot-water boiler, manufactured in 1988, with a nominal capacity of 4,433 kW and a design supply/return temperature regime of 130/90 ℃. Although the burner was reportedly replaced in 2024, this intervention did not replace the boiler body or establish the seasonal performance of the complete heating system.

An effective seasonal system efficiency of 0.55 was retained from the engineering assessment. This parameter relates the fuel-energy input to the useful heat delivered within the assessment boundary and therefore represents the combined effect of the losses included in that assessment. It was not treated as a measured combustion efficiency or as a manufacturer-rated boiler efficiency.

The reported installed capacity alone does not establish the extent of oversizing attributable to the TEB, because the boiler plant also serves infrastructure within the wider hospital complex. Similarly, annual heat demand cannot determine the required peak capacity. The present comparison therefore uses seasonal efficiency to estimate annual fuel requirements, while treating equipment sizing and service redundancy as separate design questions.

2.3 Reconstruction of a Consistent Calculation Baseline

The source assessment reports an annual useful-heat requirement of 1,297,320.38 kWh. Baseline final energy ($E_{\mathrm{f},0}$) and primary energy ($E_{\mathrm{p},0}$) were calculated as:

$$E_{\mathrm{f},0} = \frac{Q_0}{\eta_0}, E_{\mathrm{p},0} = f_{\mathrm{p}} E_{\mathrm{f},0} \tag{1}$$

where, Q_0 is the annual useful-heat requirement, $\eta_0 = 0.55$ is the baseline seasonal system efficiency, and $f_\mathrm{p} = 1.10$ is the primary-energy factor adopted in the source calculations. Final energy denotes the fuel-energy input, whereas primary energy includes the adopted upstream conversion factor.

Fuel consumption (V_0) and expenditure (C_0) were calculated from the same energy balance:

$$V_0 = \frac{E_{\mathrm{f},0}}{H_\mathrm{L}}, C_0 = p_\mathrm{f} V_0 \tag{2}$$

where, $H_\mathrm{L} = 10.56$ kWh/L is the assumed lower heating value and $p_\mathrm{f} = 1.50$ BAM/L is the fuel price.

These inputs yield a final-energy requirement of 2,358,764.33 kWh/year, a primary-energy requirement of 2,594,640.76 kWh/year, a corresponding fuel volume of 223,367.83 L/year and an annual fuel expenditure of 335,051.75 BAM. Table 1 summarises the calculation baseline.

Table 1. Building characteristics and inputs used to reconstruct the annual calculation baseline

Parameter	Symbol	Unit	Value
Usable floor area	A_f	m^2	12,947.28
Heated volume	V_h	m^3	62,434.72
External-envelope area	A_env	m^2	17,113.65
Indoor design temperature	T_in	°C	22
Operating schedule	—	h/day	24
Annual useful-heat requirement	Q_0	kWh/year	1,297,320.38
Baseline seasonal system efficiency	η_0	—	0.55
Fuel lower heating value	H_L	kWh/L	10.56
Primary-energy factor	f_p	—	1.10
Fuel price	p_f	BAM/L	1.50
Calculated annual final energy	$E_{\mathrm{f},0}$	kWh/year	2,358,764.33
Calculated annual primary energy	$E_{\mathrm{p},0}$	kWh/year	2,594,640.76
Calculated annual fuel consumption	V_0	L/year	223,367.83
Calculated annual fuel expenditure	C_0	BAM/year	335,051.75

Note: "—" = not applicable; BAM = Bosnia and Herzegovina convertible mark.

A reported annual fuel consumption of 194,510 L corresponds to an expenditure of 291,765 BAM at the assumed fuel price. These two quantities are mutually consistent at the assumed fuel price, but the reported fuel volume corresponds to only 2,054,025.60 kWh of final energy. This fuel-consumption figure is inconsistent with the useful-heat requirement and seasonal efficiency adopted in Table 1. The annual scenario comparison was reconstructed using Eqs. (1) and (2); the reported fuel volume was retained as a separate reference value rather than used as evidence of model calibration.

An operational CO_2 emission factor (g_f) was inferred from the reported energy savings and CO_2 reduction for envelope renovation:

$$g_\mathrm{f} = \frac{\Delta M_1}{\Delta E_{\mathrm{p},1}/f_\mathrm{p}} \approx 0.275735 \text{ kg } \mathrm{CO_2}/\mathrm{kWh} \tag{3}$$

The source values were $\Delta M_1 = 367{,}924.8$ kg CO_2/year and $\Delta E_{\mathrm{p},1} = 1{,}467{,}776.2$ kWh/year. This is a source-implied conversion factor, not an independently verified fuel-property measurement. Applying the unrounded factor consistently to the reconstructed baseline gives approximately 650,393.90 kg CO_2/year. Emission results were interpreted as conditional estimates of operational CO_2 reductions; embodied emissions and non-CO_2 greenhouse gases were outside the assessment boundary.

2.4 Climatic Data and Assessment Scope

The source assessment refers to climatic information for Banja Luka obtained from Meteoblue and solar-resource information from the European Commission's Photovoltaic Geographical Information System (PVGIS) [33, 34]. These sources provide climatic context for heating demand and solar availability. The available records did not identify the weather series, averaging period or collector-plane radiation used to derive the annual estimates.

The present reconstruction therefore retained the annual useful-heat requirement and useful solar-heat estimate supplied in the engineering assessment. The comparison used the reported annual heat-demand and solar-output estimates without additional hourly simulation or calibration against measured consumption.

3 Renovation Scenarios and Evaluation Methods

3.1 Definition of the Renovation Measures

3.1.1 Thermal-envelope renovation

Measure M1 comprises insulation of external walls and soffits, insulation of flat roofs, and replacement of external windows, doors and the glazed roof above the boiler room. The specified wall intervention uses 100 mm of mineral wool within an external thermal insulation composite system. Flat-roof insulation comprises 160 mm of mineral wool or extruded polystyrene, depending on the roof construction. Replacement windows use thermally broken aluminium frames and improved glazing, with a reported target thermal transmittance of approximately 1.5–1.6 W/(m^2·K).

These interventions were evaluated as one envelope package because the available scenario data report their combined effect. No separate contribution was assigned to windows, walls or roofs in the absence of component-level heat-loss calculations. Table 2 summarises the technical specification.

Table 2. Measures and principal inputs used in the scenario comparison

Measure	Technical Specification	Representation in the Annual Model	Source Investment Estimate (BAM)
M1 Thermal envelope	Wall and soffit insulation; roof insulation; replacement windows, doors and boiler-room glazing	Reduction in annual useful-heat requirement	1,844,006.20
M2 Heat supply	Replacement by a cascade-operated boiler arrangement	Seasonal system efficiency increased from 0.55 to 0.76	335,790.00
M3 Solar thermal	24 flat-plate collectors; 44 m^2 absorber area; associated storage, heat exchange, circulation and controls	Annual useful solar heat of 23,589.83 kWh	322,337.34

Note: BAM = Bosnia and Herzegovina convertible mark.

The annual useful-heat requirement after envelope renovation (Q_{env}) was reconstructed from the source value for M1 primary-energy savings:

$$Q_{\mathrm{env}} = \frac{\eta_0}{f_{\mathrm{p}}} \left(E_{\mathrm{p},0} - \Delta E_{\mathrm{p},1}\right) = 563,432.28 \text{ kWh/ year} \tag{4}$$

Accordingly, the envelope package represents a reduction of 733,888.10 kWh/year in useful-heat demand. This reconstruction reproduces the aggregate envelope effect reported in the source assessment; it is not an independent calculation from element areas, thermal transmittances and ventilation rates.

3.1.2 Boiler replacement

Measure M2 represents a replacement heat-supply arrangement with two boilers operating in cascade. The intended purpose is to match heat generation more closely to demand, with staged operation across changing load conditions. In the annual model, replacement was represented by an effective seasonal system efficiency of 0.76, compared with 0.55 for the existing system.

The efficiency change was treated as a scenario assumption rather than as a measured post-renovation outcome. It also differs from a product's nominal or standard seasonal efficiency, which is determined under specified test conditions. Manufacturer-rated performance [35] was distinguished from the effective seasonal system efficiency adopted in the annual model.

Scenarios containing M2 but excluding envelope renovation were retained to compare the annual effect of the efficiency assumption against the unrenovated heat demand. They were not regarded as confirmation that equipment selected for the renovated building could meet the unrenovated peak load. Detailed implementation requires a separate design-load calculation, confirmation of the distribution temperature regime and an assessment of standby capacity for essential hospital services.

3.1.3 Solar thermal integration

Measure M3 comprises 24 flat-plate collectors with a total absorber area of 44 m^2, together with storage, heat exchangers, pumps, controls and connecting pipework. The source assessment associates this configuration with approximately 2,500 L/day of domestic hot-water demand and an annual useful solar contribution of 23,589.83 kWh.

The reported solar fraction of approximately 40.4% implies an annual useful domestic hot-water requirement of approximately 58.39 MWh. This value was used as a consistency indicator rather than as an independently calculated hot-water demand, because inlet-water temperature, delivery temperature and the draw-off schedule were not specified in the available material.

Useful solar heat was interpreted as heat offset at the same demand boundary used for the conventional system. Its assumed annual value was held constant across scenarios containing M3. The fuel savings associated with this contribution were allowed to vary with the efficiency of the remaining conventional heat supply. This distinction is necessary because an unchanged solar contribution displaces less fuel when the auxiliary heating system is more efficient. The collector configuration was therefore evaluated as a specified engineering option, rather than presented as an optimised or independently validated design.

3.2 Scenario Construction and Investment Boundary

The three measures were combined into seven non-baseline scenarios, as shown in Table 3. S0 denotes the existing calculation baseline and was used to determine savings; it was not included as an alternative in the PSI decision matrix.

Table 3. Definition of the baseline and renovation scenarios

Scenario	M1 Envelope	M2 Boilers	M3 Solar Thermal	Description
S0	—	—	—	Existing calculation baseline
S1	✓	—	—	Envelope renovation
S2	—	✓	—	Boiler replacement
S3	—	—	✓	Solar thermal installation
S4	✓	✓	—	Envelope renovation and boiler replacement
S5	✓	—	✓	Envelope renovation and solar thermal
S6	—	✓	✓	Boiler replacement and solar thermal
S7	✓	✓	✓	All three assessed measures

Note: M = measure; "—" = not applicable; "✓" = applicable.

S7 combines the three assessed measures within the defined heating and domestic hot-water boundary.

Investment estimates were retained from the source assessment and include value-added tax. Combined-scenario investments were calculated by adding the constituent measure costs. Combined investments were calculated without adjustments for procurement discounts, shared construction activities or additional enabling works. The estimates were used on their original common price basis and were not represented as current procurement quotations.

3.3 Annual Energy Model and Interactions Between Measures

For each scenario i, the useful-heat requirement (Q_i) was defined as:

$$Q_i = \begin{cases} Q_{\text{env}}, & \text{if M1 was included} \\ Q_0, & \text{otherwise} \end{cases} \tag{5}$$

The effective seasonal system efficiency (η_i) was:

$$\eta_i = \begin{cases} 0.76, & \text{if M2 was included} \\ 0.55, & \text{otherwise} \end{cases} \tag{6}$$

Let $s_i = 1$ for scenarios containing M3 and $s_i = 0$ otherwise. Annual final energy ($E_{\text{f},i}$) and primary energy ($E_{\text{p},i}$) were then calculated as:

$$E_{\text{f},i} = \frac{Q_i - s_i Q_{\text{sol}}}{\eta_i}, E_{\text{p},i} = f_{\text{p}} E_{\text{f},i} \tag{7}$$

where, $Q_{\text{sol}} = 23{,}589.83$ kWh/year. Solar heat was deducted from useful demand before applying conventional-system efficiency. This formulation represents the reduction in fuel-derived heat rather than a direct subtraction of solar output from fuel-energy input.

Eq. (7) also prevents the effects of combined measures from being calculated by adding their standalone savings. Envelope renovation reduces the heat requirement to which the improved boiler efficiency applies. Conversely, boiler replacement reduces the amount of fuel displaced by a given useful solar contribution. These interactions were represented within the annual balance without introducing additional assumptions about hourly dispatch or part-load performance.

3.4 Performance Indicators and Criterion Dependence

Annual savings were calculated relative to S0:

$$\begin{aligned} |\Delta E_{\mathrm{p},i}| &= E_{\mathrm{p},0} - E_{\mathrm{p},i}, \\ |\Delta V_i| &= \frac{E_{\mathrm{f},0} - E_{\mathrm{f},i}}{H_{\mathrm{L}}}, \\ |\Delta M_i| &= g_{\mathrm{f}} \left(E_{\mathrm{f},0} - E_{\mathrm{f},i}\right), \\ |\Delta C_i| &= p_{\mathrm{f}} \Delta V_i \end{aligned} \tag{8}$$

where, ΔM_i denotes avoided operational CO_2 emissions and ΔC_i denotes avoided fuel expenditure. Fuel-cost savings exclude changes in maintenance, auxiliary electricity, inspection and component replacement costs.

Four criteria were retained for the reference PSI analysis: primary-energy savings, operational CO_2 reduction, investment and annual fuel-cost savings. Their directions are listed in Table 4.

Table 4. Criteria used in the reference decision matrix

Criterion	Symbol	Unit	Preferred Direction
Annual primary-energy savings	K_1	kWh/year	Higher
Annual operational CO_2 reduction	K_2	kg/year	Higher
Initial investment	K_3	BAM	Lower
Annual fuel-cost savings	K_4	BAM/year	Higher

Note: BAM = Bosnia and Herzegovina convertible mark.

Under the common fuel, emission factor and price assumptions, the three savings criteria are related by:

$$\Delta M_i = \frac{g_{\mathrm{f}}}{f_{\mathrm{p}}} \Delta E_{\mathrm{p},i}, \Delta C_i = \frac{p_{\mathrm{f}}}{H_{\mathrm{L}} f_{\mathrm{p}}} \Delta E_{\mathrm{p},i} \tag{9}$$

These indicators express the same fuel reduction in different units; including all three in an additive ranking therefore duplicates information. The four-criterion formulation was retained as a reference to the source analysis, while a reduced formulation was used to examine the effect of removing this duplication. Simple payback was excluded from both decision matrices because it is derived from investment and annual savings.

3.5 Preference Selection Index Procedure

The PSI calculation followed the original procedure and its subsequent engineering application [30, 31]. The decision matrix contained seven alternatives and four criteria. Each entry x_{ij} represented the performance of scenario i against criterion j.

Benefit and cost criteria were normalized as:

$$r_{ij} = \begin{cases} \frac{x_{ij}}{\max_i x_{ij}}, & \text{for a benefit criterion,} \\ \frac{\min_i x_{ij}}{x_{ij}}, & \text{for a cost criterion} \end{cases} \tag{10}$$

All entries used in the calculation were positive. The normalized mean ($\bar{r}_j$) and preference variation for each criterion (PV_j) were calculated as:

$$\bar{r}_j = \frac{1}{n} \sum_{i=1}^{n} r_{ij}, PV_j = \sum_{i=1}^{n} \left(r_{ij} - \bar{r}_j\right)^2 \tag{11}$$

where, $n = 7$. Preference deviations and normalized criterion weights were then obtained from:

$$\phi_j = 1 - PV_j, w_j = \frac{\phi_j}{\sum_{k=1}^{m} \phi_k} \tag{12}$$

where, $m = 4$ for the reference formulation. Preference deviations were checked before interpreting the resulting values as non-negative weights.

The overall preference score was:

$$I_i = \sum_{j=1}^{m} w_j r_{ij} \tag{13}$$

A higher score indicates a more favorable alternative within the specified criteria, normalization and weighting procedure. The weights were derived from the decision matrix and did not represent preferences elicited from hospital decision makers.

3.6 Ranking Robustness and Decision Priorities

Ranking robustness was examined separately from the accuracy of the engineering inputs. First, the reference PSI calculation used all four criteria. Second, PSI was recalculated using only primary-energy savings and investment, thereby retaining the energy–capital trade-off while removing the proportional savings criteria identified in Eq. (9).

A further comparison assigned equal weight to the two retained criteria. More generally, the effect of the investment weight was represented as:

$$J_i(\lambda) = (1-\lambda)\frac{\Delta E_{\mathrm{p},i}}{\max_k \Delta E_{\mathrm{p},k}} + \lambda \frac{\min_k C_{\mathrm{inv},k}}{C_{\mathrm{inv},i}}, 0 \le \lambda \le 1 \tag{14}$$

where, $C_{\mathrm{inv},i}$ is scenario investment. Increasing λ places greater importance on lower initial expenditure, while decreasing it prioritises larger absolute energy savings. Equal weighting corresponds to $\lambda = 0.5$. For any two alternatives, their crossover weight can be obtained by setting their scores in Eq. (14).

These comparisons distinguish a ranking produced by the source-derived weights from a decision made under another allocation of priorities. This analysis assesses sensitivity to decision priorities; uncertainty in the engineering inputs is addressed separately in Section 5.6.

3.7 Economic Comparison and Incremental Solar Investment

The simple payback period was calculated as:

$$PP_i = \frac{C_{\mathrm{inv},i}}{\Delta C_i} \tag{15}$$

For the addition of solar thermal to the envelope-and-boiler package, the relevant comparison was S7 against S4. Incremental investment, incremental fuel-cost savings and incremental simple payback were calculated as:

$$\begin{aligned} \left|\Delta C_{\mathrm{inv},7|4}\right| &= C_{\mathrm{inv},7} - C_{\mathrm{inv},4}, \\ \left|\Delta C_{\mathrm{save},7|4}\right| &= \Delta C_7 - \Delta C_4, \\ \left|PP_{7|4}\right| &= \frac{\Delta C_{\mathrm{inv},7|4}}{\Delta C_{\mathrm{save},7|4}}. \end{aligned} \tag{16}$$

The same comparison was used to calculate the additional primary-energy savings and CO_2 reductions obtained from the solar installation. This incremental assessment addresses whether the additional expenditure is justified by the additional savings, rather than attributing the financial performance of the entire package to its final component.

Fuel-price sensitivity was evaluated using prices 20% below and 20% above the reference value, with investment, heat demand and system efficiency held constant. For a positive price multiplier a,

$$\Delta C_i(a) = a\Delta C_i, PP_i(a) = \frac{PP_i}{a} \tag{17}$$

A uniform price change scales every scenario's fuel-cost saving by the same multiplier. Under the normalization in Eq. (10), this multiplier cancels, leaving the reference PSI scores unchanged. Fuel-price sensitivity therefore describes financial exposure, rather than providing an independent test of ranking robustness.

Both total and incremental payback were interpreted as undiscounted screening indicators. They do not include financing, maintenance changes, replacement expenditure, fuel-price escalation or residual value, and were not used to infer positive net present value or life-cycle economic viability.

4 Results

4.1 Energy Savings and Financial Performance

Table 5 presents the recalculated performance of the seven scenarios. The estimates were obtained from the common baseline and interaction model defined in Sections 2 and 3. Energy savings, avoided emissions and fuel-cost savings therefore refer to the same heating and domestic hot-water boundary.

Among the individual measures, envelope renovation provided the largest reduction in annual primary-energy demand. S1 saved 1,467.776 MWh/year, equivalent to 56.57% of the reconstructed baseline. Its energy savings were approximately twice those of boiler replacement and 31.1 times those of the solar installation. This result reflects the aggregate demand reduction assigned to the envelope package; the available data do not support attributing a specific share to windows, walls or roofs.

Boiler replacement required substantially less investment than envelope renovation and produced the shortest simple payback period, at 3.63 years. Its investment per unit of annual primary-energy savings was approximately

0.47 BAM/(kWh/year), compared with 1.26 BAM/(kWh/year) for S1. These ratios describe the initial capital required to obtain an annual energy saving, rather than a levelised cost of conserved energy. The favourable financial result for S2 remains conditional on the assumed seasonal efficiency and does not establish that the proposed equipment capacity would meet the unrenovated peak load.

Table 5. Recalculated energy, environmental and financial performance of the renovation scenarios

Scenario	Primary-Energy Savings (MWh/year)	CO_2 Reduction (t/year)	Investment (BAM)	Fuel-Cost Savings (BAM/year)	Simple Payback (years)
S1	1,467.776	367.925	1,844,006.20	189,537.22	9.73
S2	716.940	179.714	335,790.00	92,580.09	3.63
S3	47.180	11.826	322,337.34	6,092.41	52.91
S4	1,779.147	445.975	2,179,796.20	229,745.18	9.49
S5	1,514.956	379.751	2,166,343.54	195,629.63	11.07
S6	751.083	188.273	658,127.34	96,989.07	6.79
S7	1,813.290	454.534	2,502,133.54	234,154.16	10.69

Note: BAM = Bosnia and Herzegovina convertible mark.

The standalone solar scenario produced the smallest absolute savings and the longest simple payback. Its estimated useful solar output of 23.590 MWh/year displaced 47.180 MWh/year of primary energy when paired with the existing system efficiency. Although S3 had the lowest initial investment, its cost was only slightly below that of S2, whereas its fuel-cost savings were much smaller.

Figure 2 compares the estimated annual primary-energy savings and initial investment of the seven renovation scenarios. Scenarios incorporating envelope renovation (S1, S4, S5, and S7) achieved larger absolute energy savings than scenarios limited to boiler replacement, solar thermal installation or their combination. This advantage was accompanied by higher capital expenditure. Boiler replacement alone (S2) occupied a different position: it required substantially less investment while delivering considerably greater savings than the standalone solar installation (S3). The figure therefore illustrates the distinction between maximising absolute energy savings and limiting the initial financial commitment.

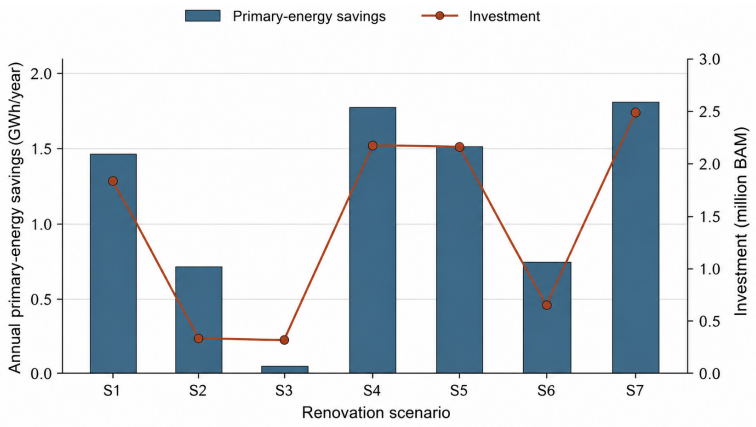


Figure 2. Estimated annual primary-energy savings and investment costs for the seven renovation scenarios

Note: BAM = Bosnia and Herzegovina convertible mark.

The comparison between S4 and S7 is particularly relevant to this trade-off. Their energy-saving bars are similar in height, whereas the investment required for S7 is appreciably higher. S4 retained 98.12% of S7's estimated primary-energy savings at 12.88% lower investment, indicating that the additional solar installation contributed relatively little to the total energy reduction compared with its capital requirement. The economic implications of this increment are examined separately in Section 4.5. Capital efficiency was assessed from the numerical ratios rather than the relative heights of the two plotted series.

4.2 Remaining Energy Demand and Operational Emissions

S7 provided the largest estimated reduction among the assessed scenarios, saving 69.89% of baseline primary energy. S4 achieved a reduction of 68.57%, while S5 achieved 58.39%. Scenarios without envelope renovation

retained a larger proportion of the baseline heat demand, despite improvements to heat supply.

Table 6 reports the energy and operational emissions remaining after each intervention. The emission values were calculated consistently from the reconstructed baseline and the source-implied emission factor described in Section 2.3.

Table 6. Remaining annual primary energy and operational CO_2 emissions

Scenario	Remaining Primary Energy (MWh/year)	Primary-Energy Reduction (%)	Remaining CO_2 Emissions (t/year)
S0	2,594.641	—	650.394
S1	1,126.865	56.57	282.469
S2	1,877.701	27.63	470.680
S3	2,547.461	1.82	638.567
S4	815.494	68.57	204.418
S5	1,079.685	58.39	270.643
S6	1,843.557	28.95	462.121
S7	781.351	69.89	195.860

Note: "—" = not applicable.

Because the model retained a common fuel, primary-energy factor and emission factor, the percentage reductions in operational CO_2 and primary energy were identical. They should consequently be interpreted as two expressions of the same calculated fuel reduction, rather than independent evidence of environmental improvement.

Using the reported usable floor area, baseline primary-energy intensity was approximately 200.40 kWh/(m^2·year), falling to 60.35 kWh/(m^2·year) under S7. These values refer only to the assessed heating and hot-water uses. They are not whole-hospital energy benchmarks or building energy-certification results.

Even under S7, approximately 781.351 MWh/year of primary energy remained within the assessment boundary. The package therefore reduced reliance on fuel oil without eliminating it. Residual fuel use prevents the assessed package from achieving zero operational CO_2 emissions.

4.3 Preference Selection Index Weights and Reference Ranking

The four-criterion PSI calculation assigned almost identical weights to primary-energy savings, CO_2 reductions and fuel-cost savings. Table 7 compares these weights with those obtained after retaining only primary-energy savings and investment.

Table 7. Criterion weights under the reference and reduced PSI formulations

Criterion	Four-Criterion PSI Weight (%)	Two-Criterion PSI Weight (%)
Primary-energy savings	29.5564	72.2874
Operational CO_2 reduction	29.5564	—
Investment	11.3309	27.7126
Fuel-cost savings	29.5564	—

Note: Small differences from 100% result from rounding; PSI = Preference Selection Index; "—" = not applicable.

In the reference formulation, the three proportional savings criteria together received 88.67% of the total weight. Their equal weights followed from their identical normalized profiles under the reconstructed model. The calculation therefore placed substantially greater aggregate emphasis on fuel reduction than on initial investment.

Removing the duplicate savings indicators increased the investment weight from 11.33% to 27.71%. This change affected the preference scores, although it did not change the ordering of the seven scenarios. Table 8 presents both PSI results and the equal-weight comparison of energy savings and investment.

Figure 3 compares the scenario preference scores obtained from the four-criterion PSI formulation, the reduced two-criterion PSI formulation and an equal-weight evaluation of primary-energy savings and investment. The two PSI formulations produced the same ordering, with S7 ranked first and S4 second. However, removing the proportional CO_2 and fuel-cost savings criteria increased the investment weight from 11.33% to 27.71% and narrowed the score difference between S7 and S4 from approximately 0.0145 to 0.0083. The unchanged ordering therefore needs to be considered alongside the change in the relative emphasis assigned to energy savings and capital expenditure.

A different result emerged when primary-energy savings and investment received equal weights. Under this setting, S2 achieved the highest score, followed by S4 and S7. Its smaller absolute energy savings were offset by its substantially lower investment requirement. The figure shows that agreement between the two PSI rankings did

not imply that the preferred scenario was independent of the weighting assumptions. Each set of bars represents scores calculated under a particular evaluation setting; changes in a scenario's score across settings do not represent changes in its physical energy performance. Section 4.4 extends this comparison by varying the investment weight continuously and identifying the points at which the highest-scoring scenario changes.

Table 8. Scenario scores and ranks under alternative evaluation settings

Scenario	Four-Criterion PSI Score	Rank	Two-Criterion PSI Score	Rank	Equal-Weight Score	Rank
S1	0.7375	4	0.6336	4	0.4921	6
S2	0.4594	5	0.5518	5	0.6777	1
S3	0.1364	7	0.2959	7	0.5130	4
S4	0.8868	2	0.7502	2	0.5645	2
S5	0.7577	3	0.6452	3	0.4921	5
S6	0.4228	6	0.4352	6	0.4520	7
S7	0.9013	1	0.7586	1	0.5644	3

Note: Ranks were calculated from unrounded scores. The displayed equal-weight scores for S1 and S5 round to the same value; PSI = Preference Selection Index.

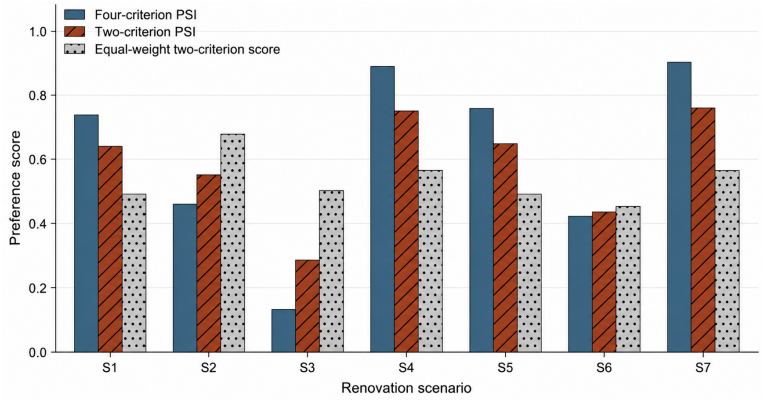


Figure 3. Scenario preference scores under the four-criterion PSI, two-criterion PSI and equal-weight two-criterion evaluations

Note: PSI = Preference Selection Index.

4.4 Investment Priorities and Ranking Changes

The investment weight, λ, was varied from 0 to 1 in the two-criterion score defined in Eq. (14), with the energy-savings weight set to $1-\lambda$. Table 9 summarises the resulting intervals over which each scenario achieved the highest score. S7 ranked first at investment weights below approximately 0.4211, S2 ranked first between approximately 0.4211 and 0.9021, and S3 ranked first above approximately 0.9021. Adjacent leading scenarios had equal scores at the crossover weights.

Table 9. Highest-scoring scenario as the investment weight changes

Investment Weight	Highest-Scoring Scenario	Interpretation Within the Specified Score
$0 \leq \lambda < 0.4211$	S7	Greater emphasis on absolute energy savings
$0.4211 < \lambda < 0.9021$	S2	Lower investment receives sufficient importance to favor boiler replacement
$0.9021 < \lambda \leq 1$	S3	The score is dominated by the lowest initial investment

Note: Adjacent scenarios have equal scores at the unrounded crossover weights, approximately 0.421123 and 0.902149.

S4 did not become the highest-scoring scenario over this continuous weighting range. Its pairwise crossover with S7 occurred at an investment weight of approximately 0.4971, but S2 already had a higher score than both alternatives at that point. A pairwise advantage over S7 therefore did not make S4 the overall preferred alternative.

This finding qualifies the interpretation of S4 as a lower-cost option. Its relevance arises from retaining a large proportion of S7's energy savings while reducing expenditure, rather than from being the highest-scoring alternative under every intermediate preference. Where decision makers require substantial demand reduction, S4 merits consideration alongside S7. Where the immediate priority is lower capital expenditure, the annual model instead favours examining S2, subject to confirmation of its technical feasibility.

The first-place position of S3 at very high investment weights also illustrates a limitation of interpreting scores without examining their construction. It reflects the preference for its slightly lower initial cost, not strong energy or financial performance.

4.5 Incremental Evaluation of Solar Thermal Integration

The direct comparison between S4 and S7 isolates the addition of solar thermal after envelope renovation and boiler replacement. Table 10 summarises this increment.

Table 10. Incremental performance of adding solar thermal to S4

Indicator	S7 Minus S4
Additional investment	322,337.34 BAM
Additional primary-energy savings	34,143.175 kWh/year
Additional operational CO_2 reduction	8,558.608 kg/year
Additional fuel-cost savings	4,408.98 BAM/year
Incremental simple payback	73.11 years

Note: S = Scenario; BAM = Bosnia and Herzegovina convertible mark.

S4 retained 98.12% of S7's primary-energy savings at an investment 12.88% below that of S7. Conversely, moving from S4 to S7 increased investment by 14.79% relative to S4, while increasing annual primary-energy savings by approximately 1.92%. The incremental solar payback was longer than the standalone solar payback of 52.91 years. The useful solar contribution was unchanged, but its avoided fuel input decreased when the auxiliary-system efficiency increased from 0.55 to 0.76. Consequently, annual fuel-cost savings attributable to solar heat fell from 6,092.41 BAM in S3 to 4,408.98 BAM when solar thermal was added to S4.

The overall payback of S7, at 10.69 years, therefore did not describe the financial return on its solar component. Most of the package's fuel-cost savings were already obtained through envelope renovation and boiler replacement. Evaluating the solar addition through the package payback alone would conceal this distinction.

4.6 Fuel-Price Sensitivity

Table 11 presents the effect of a uniform 20% decrease or increase in fuel price. The calculation held investment and physical performance constant.

Table 11. Simple payback under alternative fuel-price assumptions

Scenario or Increment	1.20 BAM/L	1.50 BAM/L	1.80 BAM/L
S2	4.53 years	3.63 years	3.02 years
S4	11.86 years	9.49 years	7.91 years
S7	13.36 years	10.69 years	8.90 years
Solar addition, S7 relative to S4	91.39 years	73.11 years	60.92 years

Note: BAM/L = Bosnia and Herzegovina convertible mark per liter.

A 20% price increase reduced simple payback by 16.67%, whereas a 20% decrease increased it by 25%. These changes followed directly from the inverse relationship between fuel price and payback under the specified assumptions. Physical energy savings and operational CO_2 reductions were unchanged.

The reference PSI ranking was also unchanged. A common price multiplier cancelled during normalization of the fuel-cost criterion, leaving its normalized values and derived weight unaffected. This invariance is a mathematical property of the formulation, rather than evidence that renovation decisions are generally insensitive to energy prices.

5 Discussion

5.1 Interpreting Energy Performance and Investment Returns

The scenario with the largest energy savings was not necessarily preferred when greater weight was assigned to investment.

S7 achieved the largest estimated energy savings among the assessed alternatives. It combined the lowest modelled useful-heat requirement with the improved heating-system efficiency and the specified solar contribution. S2, however, offered the shortest simple payback and required substantially less capital. Its limited demand reduction did not negate that financial advantage, just as S7's larger energy savings did not establish the strongest return on each component of its investment.

S4 occupied a different position. It achieved nearly the same energy savings as S7 without the solar installation, making it relevant where substantial energy reduction is required but expenditure is constrained. The weighting analysis nevertheless showed that S4 was not the highest-scoring alternative across the full range of the two-criterion score. Its practical relevance must therefore be stated in relation to a required level of energy reduction or an implementation objective, rather than described as an unconditional compromise solution.

This distinction is consistent with research that incorporates investor preferences into hospital renovation decisions [7] and examines conflicting stakeholder priorities in building retrofit selection [25]. The present case provides a numerical example of why the most comprehensive package and the preferred allocation of limited capital need not coincide.

5.2 Interactions Between Demand Reduction and Heat Supply

The interaction model explains both the reduced combined savings of envelope and boiler measures and the longer incremental payback of solar thermal. Once envelope renovation reduces useful demand, the replacement boiler acts on a smaller annual heat requirement. Once boiler efficiency improves, each unit of useful solar heat avoids less fuel. Neither effect is captured by adding the standalone savings.

Combined savings were smaller than the sum of the standalone estimates because the measures displaced overlapping fuel consumption. In this case, the combined savings were smaller than the sum of the standalone estimates because the measures acted on overlapping fuel consumption. The value of combining them lay in achieving a larger total reduction than either measure alone, while avoiding double counting.

This annual representation remains simpler than a dynamic system assessment. It does not resolve changes in return-water temperature, boiler cycling, collector utilization or storage losses. Research on heating-system retrofits identifies generation and control as important intervention areas [9], while robust solar-system design explicitly considers uncertainty in operating conditions [14]. Those mechanisms would require additional input data before their effects could be quantified for the TEB.

5.3 Solar Thermal as an Additional Investment

The solar installation contributed renewable heat and reduced conventional fuel use, but the specified investment was high relative to its estimated annual output. Its weak financial performance should therefore be attributed to the assessed configuration and cost assumptions, rather than treated as a general property of hospital solar thermal systems.

Previous hospital research has examined improvements to existing solar-and-boiler installations [12], alternative collector and storage configurations [13], and robust system sizing under uncertainty [14]. These studies involve different hot-water demands, system arrangements and climatic conditions. Their findings establish relevant design considerations but do not provide directly interchangeable payback benchmarks for the present installation.

A hypothetical 50% grant applied to the additional solar investment would reduce the owner's incremental simple payback from 73.11 to approximately 36.55 years, assuming unchanged performance and other costs. The 36.55-year value applies to the solar addition after implementation of S4.

A grant changes the expenditure borne by the owner; it does not reduce the full resource cost of the installation or improve its physical efficiency. Accordingly, the calculation alone does not establish that a subsidy is economically justified. Such a judgement would require a defined service life, replacement and maintenance costs, financing assumptions and an explicit valuation of outcomes beyond avoided fuel expenditure. This is consistent with the broader financial evaluation requirements identified in recent retrofit assessment literature [16].

5.4 Meaning and Limits of the Preference Selection Index Ranking

The reference PSI result was reproducible, but its interpretation depended on criterion selection. Primary-energy savings, avoided CO_2 emissions and fuel-cost savings were proportional under the model assumptions. Including them separately gave repeated representation to fuel reduction, with a combined weight of 88.67%.

The reduced PSI formulation removed that duplication and retained the same ranking, although the score gap between S7 and S4 decreased. This agreement provides evidence that the first-place PSI result was not solely caused by retaining three proportional savings criteria. It does not establish preference independence: S2 became first under equal weights, and the continuous analysis identified a change in the leading alternative at an investment weight of approximately 42.11%.

Studies comparing decision methods and criterion weights similarly show the importance of testing how preferences affect building-design choices [24, 26]. The present results suggest that reporting the decision matrix, weights and alternative ranking settings is more informative than presenting a single score as an objective recommendation.

Compared with the earlier hospital-renovation application of PSI [32], the present analysis explicitly examines proportional criteria and the incremental return from adding solar thermal after envelope and boiler improvements. This comparison explains why the highest PSI score and the strongest financial case for an additional measure need not coincide.

5.5 Implementation Conditions

The engineering sequence should allow the heat-supply design to respond to the demand expected after envelope renovation. Envelope specification and load assessment should therefore inform final boiler selection before equipment is ordered. This does not require every construction activity to occur sequentially; it requires the design decisions to use a consistent post-renovation demand basis.

The annual model cannot determine whether two proposed boilers provide sufficient peak capacity or standby capability. This is particularly important for S2 and S6, which retain the unrenovated envelope. Their energy and payback estimates describe the assumed efficiency improvement, while their implementation remains conditional on a separate capacity assessment.

Because the building supports hospital services, construction planning must also address temporary heat supply, commissioning arrangements and access to functioning technical areas. These requirements were not assigned numerical scores in the present comparison. Consequently, none of the rankings establishes that a scenario is preferable in terms of disruption, reliability or resilience.

For S4 and S7, the planning decision is whether the additional solar investment is warranted under the institution's objectives and financing conditions. Provision for future solar installation may be considered during roof works, but future installation should remain subject to an updated demand, cost and system-design assessment.

5.6 Limitations and Applicability

The first limitation concerns the calculation baseline. The reported fuel volume does not reconcile with the source useful-heat requirement, system efficiency and fuel heating value. The reconstructed baseline provides a consistent basis for comparing scenarios, but it has not been calibrated against a fuel record with a confirmed year and supply boundary. Absolute savings should therefore be treated as engineering estimates pending reconciliation of those records.

Operational CO_2 results have an additional qualification: the emission factor was inferred from the source scenario values rather than verified independently. Their proportional relationship with fuel savings is established by the model, but the absolute emissions require confirmation of the fuel-specific factor. The assessment also excludes embodied impacts, component replacement and other life-cycle stages; it therefore supports conclusions about operational energy and investment trade-offs, rather than comprehensive sustainability performance [17–20].

The physical model uses a single thermal zone, fixed seasonal efficiencies and a fixed annual solar contribution. It does not independently reproduce component heat losses, hourly loads or collector performance. Uncertainty in these inputs was not quantified through probabilistic simulation. The weighting and price analyses must consequently be distinguished from an uncertainty analysis of physical performance.

Finally, the investment values retain the source price basis, while financial savings include avoided fuel expenditure only. Changes in maintenance, auxiliary electricity and replacement costs could alter the comparison. Further assessment should prioritise reconciliation of the baseline, verification of design loads and equipment specifications, and a life-cycle financial comparison using updated costs. If measures are implemented, weather-adjusted fuel monitoring would provide evidence against which the predicted savings could be evaluated.

6 Conclusions

This study reconstructed an annual energy comparison for seven renovation scenarios in a hospital support building and examined how investment priorities affected their interpretation. The assessment accounted for the interaction between envelope demand reduction, improved heating-system efficiency and useful solar heat, rather than adding standalone savings.

The scenario containing all three measures, S7, provided the largest estimated primary-energy savings, at 1.813 GWh/year or 69.89% of the reconstructed baseline. It also ranked first under both the four-criterion and reduced two-criterion PSI formulations. However, the reference model assigned 88.67% of its total weight to three proportional savings criteria, and the leading alternative changed to S2 when primary-energy savings and investment were weighted equally. The PSI result therefore represents a preference conditional on the evaluation settings, rather than an unconditional investment optimum.

Envelope renovation combined with boiler replacement, S4, retained 98.12% of S7's primary-energy savings at 12.88% lower investment. Adding solar thermal required 322,337.34 BAM and produced an estimated additional fuel-cost saving of 4,408.98 BAM/year, corresponding to an incremental simple payback of 73.11 years. The package payback of S7 consequently did not represent the return on its solar component.

The principal contribution is the distinction between maximum energy savings, ranking-based preference and marginal investment returns within an interaction-aware renovation assessment. For the analysed options, this distinction helps explain why a lower-cost package may warrant consideration even when it does not achieve the highest PSI score. The numerical findings remain conditional on the reconstructed baseline, assumed system efficiencies, source-implied emission factor and investment estimates. Confirmation of these inputs is required before the results can support a final implementation decision.

Author Contributions

Conceptualization, T.R., S.V., and D.V.; methodology, T.R., S.V., and D.V.; formal analysis, T.R., S.V., and D.V.; investigation, T.R., S.V., and D.V.; data curation, T.R., S.V., and D.V.; writing—original draft preparation, T.R., S.V., and D.V.; writing—review and editing, T.R., S.V., D.V., M.K., J.Š., and G.K.; supervision, M.K., J.Š., and G.K.; validation, M.K., J.Š., and G.K. All authors have read and agreed to the published version of the manuscript.

Funding

This work is funded by the European Research Executive Agency (REA) under the project "ENPOWER—Enhancing Scientific Capacity for Energy Poverty (Grant No.: 101160253—ENPOWER—HORIZON-WIDERA-2023-ACCESS-02).

Data Availability

Data used in this research are accessible upon request. For access, please contact the first author via tatjana.ruzicic@gmail.com.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

AI tools were used for proofreading and improving linguistic quality. All corrections were manually reviewed and approved by the authors, who take full responsibility for the content.

References

[1] D. Bjelland, J. Brozovsky, and B. D. Hrynyszyn, "Systematic review: Upscaling energy retrofitting to the multi-building level," *Renew. Sustain. Energy Rev.*, vol. 198, p. 114402, 2024. https://doi.org/10.1016/j.rser.2024.114402

[2] K. Alexakis, V. Benekis, P. Kokkinakos, and D. Askounis, "Genetic algorithm-based multi-objective optimisation for energy-efficient building retrofitting: A systematic review," *Energy Build.*, vol. 328, p. 115216, 2024. https://doi.org/10.1016/j.enbuild.2024.115216

[3] D. R. Liyanage, K. Hewage, S. A. Hussain, F. Razi, and R. Sadiq, "Climate adaptation of existing buildings: A critical review on planning energy retrofit strategies for future climate," *Renew. Sustain. Energy Rev.*, vol. 199, p. 114476, 2024. https://doi.org/10.1016/j.rser.2024.114476

[4] M. Psillaki, N. Apostolopoulos, I. Makris, P. Liargovas, S. Apostolopoulos, P. Dimitrakopoulos, and G. Sklias, "Hospitals' energy efficiency in the perspective of saving resources and providing quality services through technological options: A systematic literature review," *Energies*, vol. 16, p. 755, 2023. https://doi.org/10.3390/en16020755

[5] C. H. Radha, "Retrofitting for improving indoor air quality and energy efficiency in the hospital building," *Sustainability*, vol. 15, p. 3464, 2023. https://doi.org/10.3390/su15043464

[6] A. Amamou, S. Blenkinsop, C. Winter, and O. Heidrich, "Net zero in healthcare buildings: Lessons from assessing the strategies of 214 NHS trusts in England," *Build. Environ.*, vol. 278, p. 112966, 2025. https://doi.org/10.1016/j.buildenv.2025.112966

[7] Y. Shi, R. Wang, and P. Chen, "Multi-criteria decision-making approach for energy-efficient renovation strategies in hospital wards: Balancing energy, economic, and thermal comfort," *Energy Build.*, vol. 298, p. 113575, 2023. https://doi.org/10.1016/j.enbuild.2023.113575

[8] Y. Shi and P. Chen, "Energy retrofitting of hospital buildings considering climate change: An approach integrating automated machine learning with NSGA-III for multi-objective optimization," *Energy Build.*, vol. 319, p. 114571, 2024. https://doi.org/10.1016/j.enbuild.2024.114571

[9] M. Krajčík, M. Arıcı, and Z. Ma, "Trends in research of heating, ventilation and air conditioning and hot water systems in building retrofits: Integration of review studies," *J. Build. Eng.*, vol. 76, p. 107426, 2023. https://doi.org/10.1016/j.jobe.2023.107426

[10] N. Banti and D. A. Krawczyk, "Integrating energy simulations and analytical hierarchy process procedure in multi-criteria evaluation of heating systems for industrial buildings," *J. Build. Eng.*, vol. 95, p. 110203, 2024. https://doi.org/10.1016/j.jobe.2024.110203

[11] C. Graf, P. Pärisch, A. Marszal-Pomianowska, M. Frandsen, B. Bendinger, and A. Cadenbach, "Domestic hot water systems in well-insulated residential buildings: A comparative simulation study on efficiency and hygiene challenges," *Energy*, vol. 313, p. 133587, 2024. https://doi.org/10.1016/j.energy.2024.133587

[12] A. Atienza-Márquez, F. D. Muñoz, F. F. Hernández, and J. M. C. López, "Domestic hot water production system in a hospital: Energy audit and evaluation of measures to boost the solar contribution," *Energy*, vol. 261, p. 125275, 2022. https://doi.org/10.1016/j.energy.2022.125275

[13] N. Matera, D. Mazzeo, and P. M. Congedo, "Energy-sustainable hospitals: Integration of a novel compound parabolic concentrator system with two storage tanks for domestic hot water production at high and low temperatures," *Appl. Therm. Eng.*, vol. 221, p. 119773, 2022. https://doi.org/10.1016/j.applthermaleng.2022.119773

[14] A. Atienza-Márquez, F. Domínguez-Muñoz, F. F. Hernández, and J. M. C. López, "Solar thermal hot water system in hospitals: Robust design methodology considering uncertainties," *Renew. Energy*, vol. 234, p. 121163, 2024. https://doi.org/10.1016/j.renene.2024.121163

[15] D. Kadrić, A. Aganovic, S. Martinović, N. Delalić, and B. Delalić-Gurda, "Cost-related analysis of implementing energy-efficient retrofit measures in the residential building sector of a middle-income country—A case study of Bosnia and Herzegovina," *Energy Build.*, vol. 257, p. 111765, 2022. https://doi.org/10.1016/j.enbuild.2021.111765

[16] M. D. Papangelopoulou, K. Alexakis, and D. Askounis, "Assessment methods for building energy retrofits with emphasis on financial evaluation: A systematic literature review," *Buildings*, vol. 15, p. 2562, 2025. https://doi.org/10.3390/buildings15142562

[17] M. Caruso, M. Buttazzoni, C. Passoni, S. Labò, A. Marini, and R. Pinho, "An updated multi-criteria decision-making method for the sustainable renovation of buildings including environmental, economic and social life-cycle metrics," *J. Build. Eng.*, vol. 98, p. 110967, 2024. https://doi.org/10.1016/j.jobe.2024.110967

[18] K. Theilig, B. Lourenço, R. Reitberger, and W. Lang, "Life cycle assessment and multi-criteria decision-making for sustainable building parts: Criteria, methods, and application," *Int. J. Life Cycle Assess.*, vol. 29, pp. 1965–1991, 2024. https://doi.org/10.1007/s11367-024-02331-9

[19] K. Slavkovic and A. Stephan, "Dynamic life cycle assessment of buildings and building stocks—A review," *Renew. Sustain. Energy Rev.*, vol. 212, p. 115262, 2025. https://doi.org/10.1016/j.rser.2024.115262

[20] E. Marsh, S. Allen, and L. Hattam, "Tackling uncertainty in life cycle assessments for the built environment: A review," *Build. Environ.*, vol. 231, p. 109941, 2022. https://doi.org/10.1016/j.buildenv.2022.109941

[21] P. Villalba, A. J. Sánchez-Garrido, and V. Yepes, "A review of multi-criteria decision-making methods for building assessment, selection, and retrofit," *J. Civ. Eng. Manag.*, vol. 30, pp. 465–480, 2024. https://doi.org/10.3846/jcem.2024.21621

[22] J. A. P. Amorocho and T. Hartmann, "A multi-criteria decision-making framework for residential building renovation using pairwise comparison and TOPSIS methods," *J. Build. Eng.*, vol. 53, p. 104596, 2022. https://doi.org/10.1016/j.jobe.2022.104596

[23] Z. Han, X. Li, J. Sun, M. Wang, and G. Liu, "An interactive multi-criteria decision-making method for building performance design," *Energy Build.*, vol. 282, p. 112793, 2023. https://doi.org/10.1016/j.enbuild.2023.112793

[24] F. Karakas, D. Grassie, Y. Schwartz, J. Dong, E. Bagkeris, D. Mumovic, J. Milner, Z. Chalabi, and A. Mavrogianni, "A multi-criteria decision analysis framework to determine the optimal combination of energy efficiency and indoor air quality schemes for English school classrooms," *Energy Build.*, vol. 295, p. 113293, 2023. https://doi.org/10.1016/j.enbuild.2023.113293

[25] D. D'Agostino, F. De Falco, F. Minelli, and F. Minichiello, "New robust multi-criteria decision-making framework for thermal insulation of buildings under conflicting stakeholder interests," *Appl. Energy*, vol. 376, p. 124262, 2024. https://doi.org/10.1016/j.apenergy.2024.124262

[26] A. P. Rachman, C. Ichwania, R. A. Mangkuto, J. Pradipta, M. D. Koerniawan, and J. Sarwono, "Comparison of multi-criteria decision-making methods for selection of optimum passive design strategy," *Energy Build.*, vol. 314, p. 114285, 2024. https://doi.org/10.1016/j.enbuild.2024.114285

[27] P. Shen, "Building retrofit optimization considering future climate and decision-making under various mindsets," *J. Build. Eng.*, vol. 96, p. 110422, 2024. https://doi.org/10.1016/j.jobe.2024.110422

[28] D. Wu, T. Zhang, J. Zhang, H. Lv, C. Yue, and M. Fu, "Sensitivity analysis and multiobjective optimization for rural house retrofitting considering construction and occupant behavior uncertainty: A case study of Jiaxian, China," *Appl. Energy*, vol. 360, p. 122835, 2024. https://doi.org/10.1016/j.apenergy.2024.122835

[29] P. Shen, Y. Li, X. Gao, Y. Zheng, P. Huang, A. Lu, W. Gu, and S. Chen, "Recent progress in building energy retrofit analysis under changing future climate: A review," *Appl. Energy*, vol. 383, p. 125441, 2025. https://doi.org/10.1016/j.apenergy.2025.125441

[30] K. Maniya and M. Bhatt, "A selection of material using a novel type decision-making method: Preference selection index method," *Mater. Des.*, vol. 31, pp. 1785–1789, 2010. https://doi.org/10.1016/j.matdes.2009.11.020

[31] R. Attri and S. Grover, "Application of preference selection index method for decision making over the design stage of production system life cycle," *J. King Saud Univ. Eng. Sci.*, vol. 27, pp. 207–216, 2015. https://doi.org/10.1016/j.jksues.2013.06.003

[32] D. Kadric, R. Blazevic, H. Bajric, and E. Kadric, "Evaluation of energy renovation measures for hospital buildings using the PSI method," *Eng. Technol. Appl. Sci. Res.*, vol. 14, pp. 12 753–12 758, 2024. https://doi.org/10.48084/etasr.6558

[33] Meteoblue, "Weather Banja Luka." https://www.meteoblue.com/en/weather/week/banja-luka_bosnia-and-herzegovina_3204541

[34] European Commission, "Photovoltaic geographical information system." https://re.jrc.ec.europa.eu/pvg_tools/en/

[35] Viessmann Climate Solutions SE, *System solutions for industry, commerce and local authorities*, Viessmann, Allendorf, Germany, 2022. https://www.viessmann.id/content/dam/public-brands/sea-master/downloads/Wegweiser%20System%20solutions%20for%20industry,%20commerce%20and%20local%20authorities%2006-2022_EN.pdf